RAKESH BORRA

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OBJECTIVE

MS Computer Science student with hands-on experience in machine learning, computer vision, language model applications, and Python-based AI workflows, seeking entry-level AI Application Engineer and Machine Learning Engineer opportunities.

EDUCATION

University of Massachusetts Boston, MA, USA **Jan 2025 – Present**
Master of Science in Computer Science

Indian Institute of Information Technology Sricity, India
Bachelor of Technology in Computer Science and Engineering

SKILLS

- **Programming & Data:** Python, SQL, NumPy, Pandas
- **Machine Learning:** Classification, Model Evaluation, TensorFlow, Keras, CNNs
- **Computer Vision & LLMs:** OpenCV, YOLO, Ollama, Prompt Engineering
- **Development Tools:** Git, GitHub, Streamlit, FastAPI, pytest

EXPERIENCE

Graduate Teaching Assistant – PHYS 113 **Sep 2026 – Present**
University of Massachusetts Boston

- Evaluate quantitative assignments and exams, identify errors, and provide consistent, detailed feedback.
- Explain technical problem-solving methods during office hours and coordinate grading with instructional staff.

Dining Services Associate **Jan 2026 – Aug 2026**
Sodexo – University of Massachusetts Boston

- Collaborated with team members to support time-sensitive operations and provide responsive service in a fast-paced environment.

PROJECTS

Clinical Trial AI – Protocol Analysis Platform (LLMs / Python / Streamlit)

- Collaborated in a team of three to build a Python application for extracting and structuring information from clinical trial PDFs using PyMuPDF, OCR, and local LLMs.
- Implemented source-page verification and automated testing workflows to support traceable protocol summaries and identify missing information.

Real-Time Smart Traffic Monitoring and Congestion Prediction System (YOLO / OpenCV / FastAPI)

- Applied YOLO and OpenCV to vehicle detection, lane-wise counting, and traffic congestion analysis from live video streams.
- Integrated computer vision outputs with FastAPI and React dashboards for traffic monitoring and density prediction.

Filter Pruning in Deep Neural Networks (CNN / Model Optimization)

- Applied structured filter pruning to convolutional neural networks to reduce computational requirements and inference latency while preserving model accuracy.
- Evaluated pruning strategies using statistical analysis and model performance comparisons, reporting up to 2x speedup.

Speech Emotion Detection (Python / Librosa / Keras)

- Developed a machine learning workflow for speech emotion recognition using Python, Librosa, and Keras.
- Processed audio features and applied neural network-based classification to identify emotional patterns in speech.

CERTIFICATIONS

The Complete Data Structures and Algorithms Course in Python – Udemy | View Certificate